

Day 6: Generative Modeling

- **Generative Modeling** — Likelihoods, KL, ELBO, model families
- 3 hours lecture + practical
- Slides, notes, and code on the course site

- What Is Generative Modeling?
- Likelihood & Divergences
- VAEs & GANs (Context)
- Building Blocks for Week 2

What Is Generative Modeling?

Discriminative vs Generative

- Discriminative: model $p(y|x)$
- Generative: model $p(x)$ or $p(x|z)$
- Sample generation, density estimation, editing

Generative AI with Stochastic Differential Equations

An introduction to flow and diffusion models

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What Is Generative Modeling? — Discriminative vs Generative (source: course materials)

Explicit vs Implicit

- Explicit likelihood: VAEs, flows, autoregressive
- Implicit: GANs (no tractable $p(x)$)
- Tradeoffs: training stability, evaluation

Generative AI - A new generation of AI systems



Artistic Images



Realistic Videos



Draft Texts

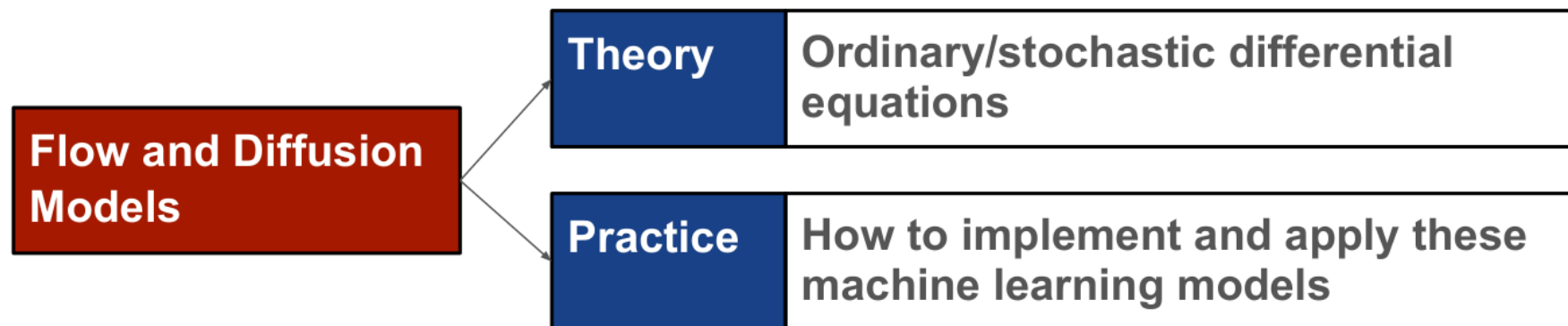
These systems are “creative”: they generate new objects.

This class teaches you algorithms to generate objects.

What Is Generative Modeling? — Explicit vs Implicit (source: course materials)

- $p(x) = \int p(x|z)p(z)dz$
- Latent z captures factors of variation
- Posterior $p(z|x)$ usually intractable

This Class: Theory and Practice of Flow/Diffusion Models



The goal of this class to teach you:

1. Flow and diffusion models from *first principles*.
2. The *minimal* but necessary amount of mathematics for 1.
3. How to implement and apply these algorithms.

What Is Generative Modeling? — Latent Variable Models (source: course materials)

- Week 2: flows, diffusion, autoregressive LLMs
- Each defines different tractability / quality

Class Overview

- **Lecture 1 (today):**
 - From Generation to Sampling: Formalize “generating an image/etc.”
 - Construct Flow and Diffusion Models
- **Lecture 2 - Flow Matching:** Training algorithm.
- **Lecture 3 - Score Matching, Guidance:** How to condition on a prompt.
- **Lecture 4 - Build Image Generators:** Network architectures + Latent spaces
- **Lecture 5 - Advanced Topics:** Discrete diffusion models, distilled models

What Is Generative Modeling? — Model Families Roadmap (source: course materials)

Likelihood & Divergences

Maximum Likelihood

- $\theta^* = \arg \max_{\theta} \sum_i \log p_{\theta}(x_i)$
- Equivalent to minimizing cross-entropy
- MLE is consistent under mild conditions

Section 1:

From Generation to Sampling

Goal: Formalize what it means to “generate” something.

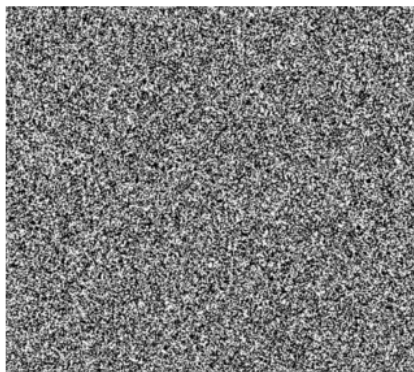
Likelihood & Divergences — Maximum Likelihood (source: course materials)

KL Divergence

- $D_{\text{KL}}(q \parallel p) = \mathbb{E}_q[\log q - \log p]$
- Non-negative; zero iff $q = p$
- Not symmetric — direction matters

What does it mean to successfully generate something?

Prompt: "A picture of a dog"



Useless

<

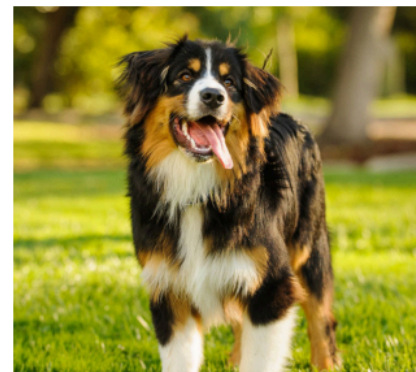
Bad

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Wrong animal

<

Great!



These are subjective statements - Can we formalize this?

Likelihood & Divergences — KL Divergence (source: course materials)

Forward vs Reverse KL

- Forward KL: mode-covering (mean-field)
- Reverse KL: mode-seeking
- VAE uses reverse KL to approximate posterior

Generation as sampling from the data distribution

Data distribution: Distribution of objects that we want to generate:

 p_{data}

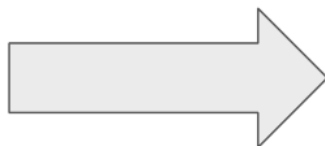
Probability density:

$$p_{\text{data}} : \mathbb{R}^d \rightarrow \mathbb{R}_{\geq 0},$$
$$z \mapsto p_{\text{data}}(z)$$

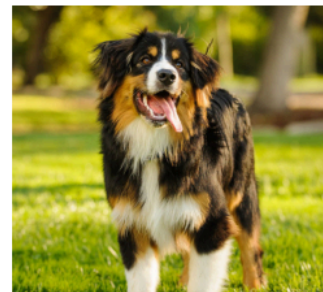
Note: We don't know the probability density!

Generation means sampling the data distribution:

$$z \sim p_{\text{data}}$$



$$z =$$



Likelihood & Divergences — Forward vs Reverse KL (source: course materials)

Evidence Lower Bound

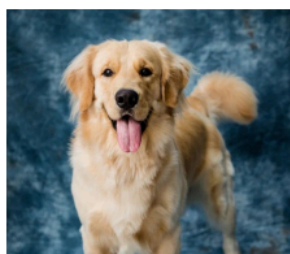
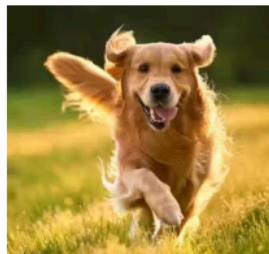
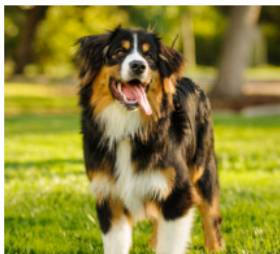
- $\log p(x) \geq \mathbb{E}_{q(z|x)}[\log p(x|z)] - D_{\text{KL}}(q(z|x) \parallel p(z))$
- ELBO tight when $q = p(z|x)$
- Reparameterization trick for gradients

Conditional Generation allows us to condition on prompts

Data distribution p_{data}

Fixed prompt

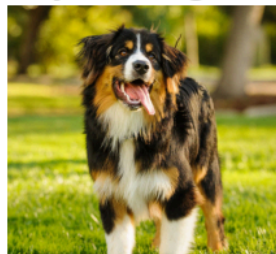
“Dog”



Conditional data distribution $p_{\text{data}}(\cdot|y)$

Condition variable: y

y = “Dog”



y = “Cat”



y = “Landscape”



Conditional generation means sampling the conditional data distribution:

$$z \sim p_{\text{data}}(\cdot|y)$$

We will first focus on unconditional generation and then learn how to translate an unconditional model to a conditional one.

Likelihood & Divergences — Evidence Lower Bound (source: course materials)

VAEs & GANs (Context)

- Encoder $q_{\varphi}(z|x)$, decoder $p_{\theta}(x|z)$
- Train by maximizing ELBO
- Blurry samples — Gaussian assumption

Summary

- **Objects to Generate:** We focus on vectors z representing data objects (e.g., images, videos)
- **Data distribution:** Distribution that places higher probability to objects that we consider “good”.
- **Generation as sampling:** generate an object = sampling from the data distribution
- **Dataset:** Finite number of samples from the data distribution used for training
- **Conditional Generation:** Condition on label y and sample from the conditional data distribution
- **Generative Model:** Train a model to transform samples from a simple (e.g., Gaussian) distribution into the data distribution.

VAEs & GANs (Context) — Variational Autoencoder (source: course materials)

- Min-max game: generator vs discriminator
- No explicit likelihood
- Mode collapse and training instability

Section 2:

Flow and Diffusion Models

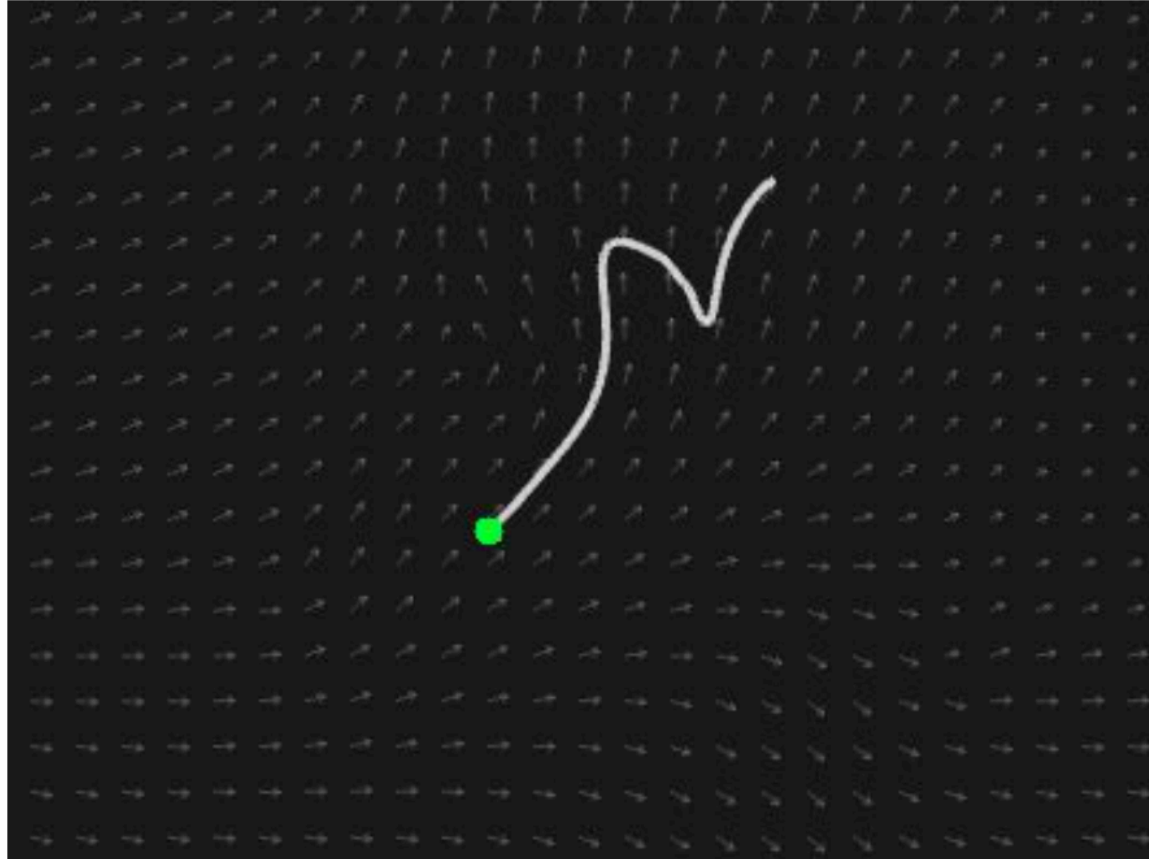
Goal: Understand differential equations and how we can build generative models with them.

VAEs & GANs (Context) — GAN Objective (source: course materials)

Evaluation Challenges

- FID, IS for images; human eval for text
- Likelihood can misalign with sample quality
- Coverage vs fidelity

ODE Trajectory - Example



Source: <https://mariogemoll.com/flow-matching>

VAEs & GANs (Context) — Evaluation Challenges (source: course materials)

- Diffusion dominates image generation
- Autoregressive dominates language
- Hybrid and unified models emerging

Existence and Uniqueness Theorem ODEs

Theorem (Picard–Lindelöf theorem): If the vector field $u_t(x)$ is continuously differentiable with bounded derivatives, then a *unique* solution to the ODE

$$X_0 = x_0, \quad \frac{d}{dt}X_t = u_t(X_t)$$

exists. In other words, a flow map exists. More generally, this is true if the vector field is **Lipschitz**.

Key takeaway: In the cases of practical interest for machine learning, **unique solutions to ODE/flows exist.**

Math class: Construct solutions via Picard-Iteration

VAEs & GANs (Context) — Modern Landscape (source: course materials)

Building Blocks for Week 2

- Score: $\nabla_x \log p(x)$
- Score matching and denoising connections
- Foundation for diffusion (Days 7–8)

- Invertible maps with tractable Jacobian
- $\log p(x) = \log p(z) + \log |\det \partial_{\frac{f}{\partial}} z|$
- Day 7 training details

Autoregressive Factorization

- $\log p(x) = \sum_t \log p(x_t \mid x_{<t})$
- Causal masks enforce ordering
- GPT family (Days 9–10)

- Day 6: **Generative Modeling**
- Likelihoods, KL, ELBO, model families
- Questions welcome — practical follows

Questions?

Practical session follows — see course site.